PROJECT DOCUMENTATION

Student Performance Analytics System

Python with AI | EWB Courses

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Course / Assignment	Python with AI – Python Project Assignment

1. Project Title

Student Performance Analytics System

2. Objective

The objective of this project is to analyze student academic performance using Python. The program processes student marks to calculate totals and averages, assign grades, determine pass/fail status, compare subject and department performance, identify top performers, and highlight students who may need academic support.

3. Technologies Used

- **Python** – Core programming language used to implement the student performance analysis system.
- **Pandas** – Reads the CSV dataset, organizes records in a DataFrame, filters records, groups data, calculates statistics, and exports the analysis report.
- **NumPy** – Converts subject marks into a NumPy array and calculates each student's total and average marks using np.sum() and np.mean().
- **Matplotlib** – Displays a bar chart of average marks by subject.
- **CSV** – Stores the student dataset and the generated analysis report.

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4. Dataset Description

The project uses a sample CSV file named Student.csv. It contains 20 student records and the following columns:

- Student_ID – student identifier.
- Name – student name.
- Department – department associated with the student.
- Python – marks in Python.
- Statistics – marks in Statistics.
- Mathematics – marks in Mathematics.
- Attendance – attendance value recorded for the student.

The dataset contains sample student records created for demonstration and learning purposes. The current analysis focuses on the three subject-mark columns: Python, Statistics, and Mathematics. Although attendance information is included in the dataset, it is not currently used in the performance calculations.

5. Implementation

1. **Load the data:** Pandas reads Student.csv into a DataFrame.
2. **Calculate totals and averages:** The program converts the Python, Statistics, and Mathematics marks into a NumPy array. It uses `np.sum()` to calculate each student's total marks and `np.mean()` to calculate the average marks.
3. **Assign grades:** A reusable calculate_grade() function assigns A for averages of 90 or above, B for 80–89.99, C for 70–79.99, D for 60–69.99, and F below 60.
4. **Determine pass/fail:** A calculate_result() function marks a student as Pass only if the student scores at least 40 in all three subjects; otherwise the result is Fail.
5. **Analyze performance:** The program identifies the top five students, class average, highest and lowest student averages, pass/fail counts, subject averages, best and weakest subjects, department averages, and students with averages below 60.
6. **Check data quality:** Pandas checks for missing values and duplicate rows.
7. **Export and visualize:** The processed DataFrame is saved as student_analysis_report.csv. Matplotlib displays a bar chart comparing average marks in the three subjects.
8. **NumPy Usage:** NumPy is used to convert the subject marks into a NumPy array and calculate each student's total and average marks using `np.sum()` and `np.mean()`. This demonstrates numerical calculations using NumPy alongside Pandas for data analysis.

6. Key Features

- Student-wise total marks, average marks, grades, and pass/fail results.
- Top five performing students.
- Overall class statistics: class average, highest average, lowest average, pass count, and fail count.
- Subject-wise average marks and identification of the best and weakest subject.
- Department-wise average performance.
- Identification of students with average marks below 60.
- Missing-value and duplicate-row checks.
- CSV export of the analyzed dataset.
- Bar chart of average marks by subject.

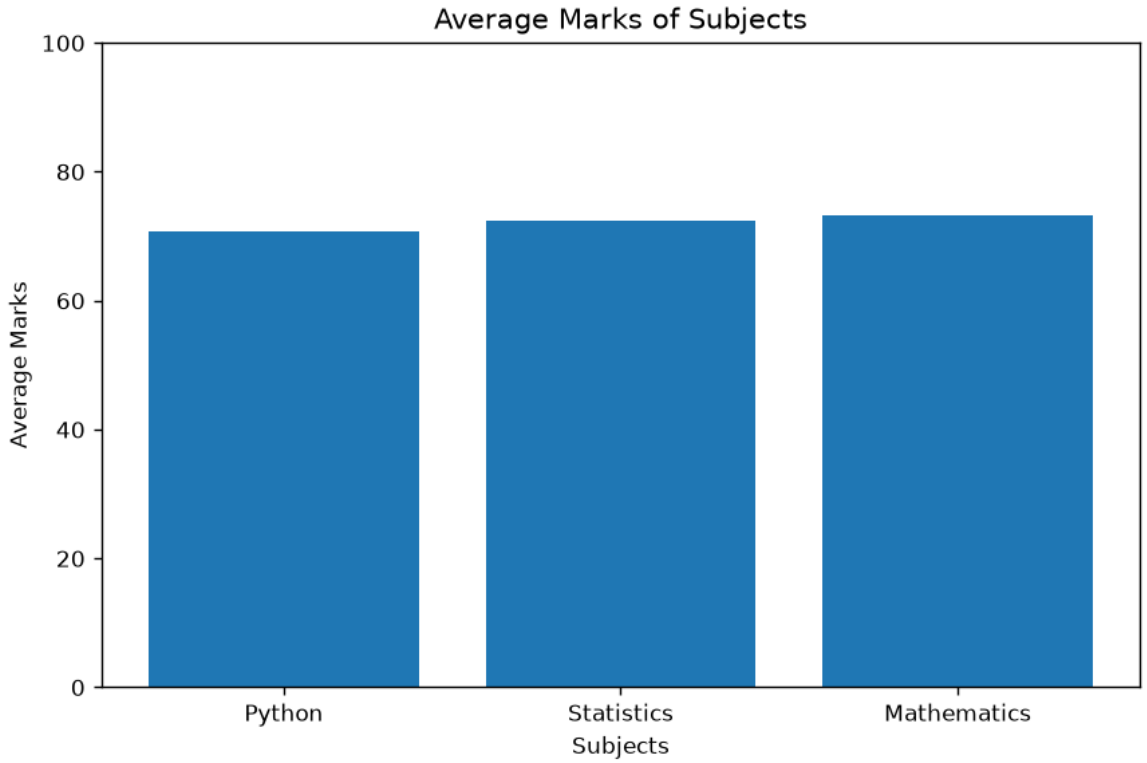
7. Output / Screenshots

When main.py runs successfully, it prints the student data and calculated results in the terminal, saves the analysis to student_analysis_report.csv, and displays a bar chart of subject-wise average marks.

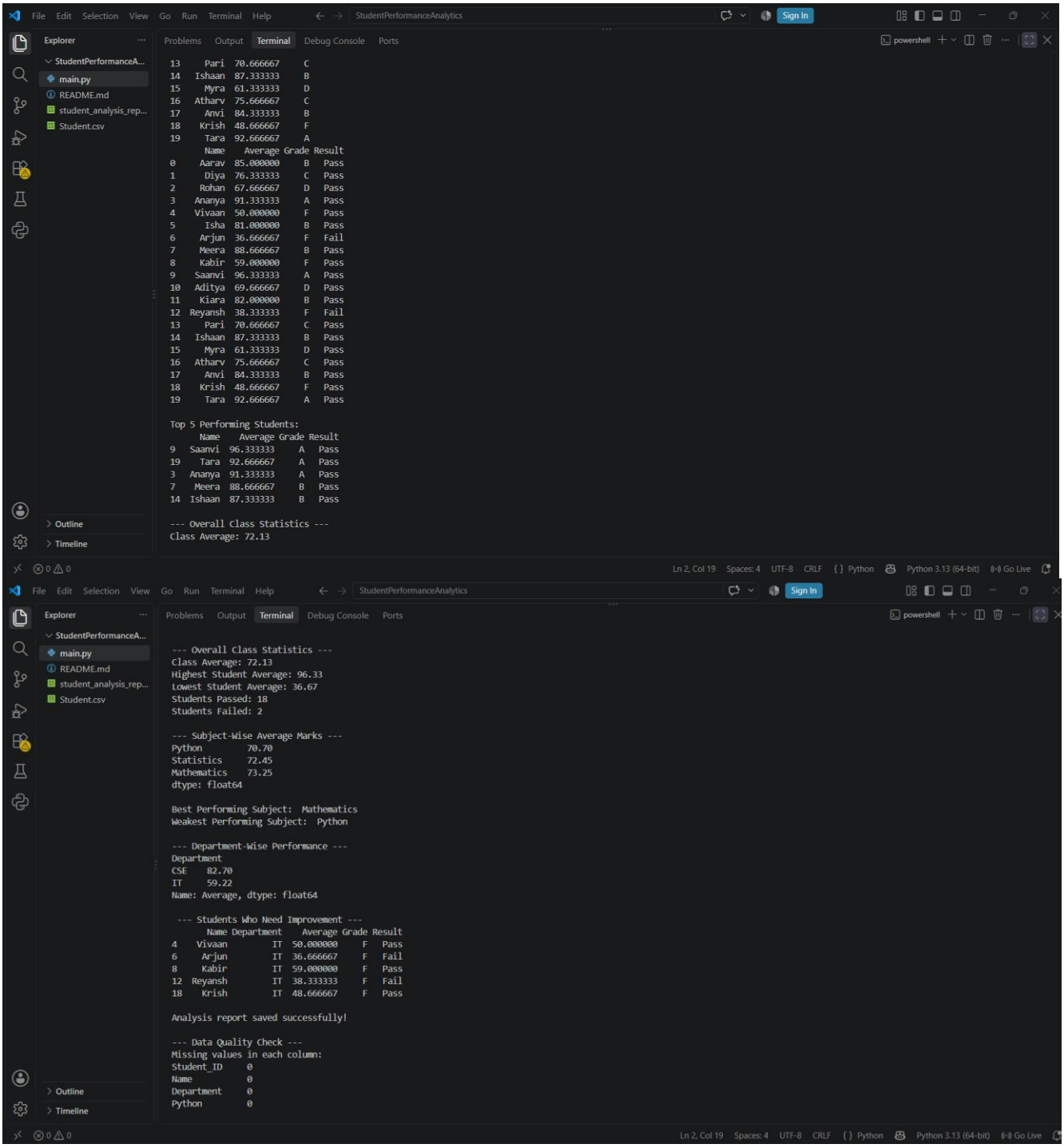
Screenshots :

Figure 1: Student Performance Analysis Outputs.

Figure 1




```
PS E:\StudentPerformanceAnalytics> py main.py
    Student_ID     Name Department  Python  Statistics  Mathematics  Attendance  Total    Average
0          101    Aarav        CSE      85          78           92          95    255  85.000000
1          102     Diya        CSE      72          81           76          88    229  76.333333
2          103    Rohan         IT      65          70           68          82    203  67.666667
3          104   Ananya        CSE      91          89           94          97    274  91.333333
4          105   Vivaan         IT      45          55           50          68    150  50.000000
5          106     Isha        CSE      78          85           80          91    243  81.000000
6          107    Arjun         IT      32          40           38          60    110  36.666667
7          108    Meera        CSE      88          92           86          96    266  88.666667
8          109    Kabir         IT      56          62           59          75    177  59.000000
9          110   Saanvi        CSE      95          96           98          99    289  96.333333
10         111   Aditya         IT      70          66           73          84    209  69.666667
11         112    Kiara        CSE      82          79           85          90    246  82.000000
12         113  Reyansh         IT      38          35           42          58    115  38.333333
13         114     Pari        CSE      67          74           71          86    212  70.666667
14         115   Ishaan         IT      90          84           88          93    262  87.333333
15         116     Myra        CSE      59          61           64          72    184  61.333333
16         117   Atharv         IT      76          72           79          87    227  75.666667
17         118     Anvi        CSE      84          88           81          94    253  84.333333
18         119    Krish         IT      48          52           46          65    146  48.666667
19         120     Tara        CSE      93          90           95          98    278  92.666667
      Name    Average Grade
0    Aarav  85.000000     B
1     Diya  76.333333     C
2    Rohan  67.666667     D
3   Ananya  91.333333     A
4   Vivaan  50.000000     F
5     Isha  81.000000     B
6    Arjun  36.666667     F
7    Meera  88.666667     B
8    Kabir  59.000000     F
9   Saanvi  96.333333     A
10  Aditya  69.666667     D
11   Kiara  82.000000     B
12 Reyansh  38.333333     F
13    Pari  70.666667     C
14  Ishaan  87.333333     B
15    Myra  61.333333     D
```


File Edit Selection View Go Run Terminal Help
StudentPerformanceAnalytics
Sign In
Explorer
StudentPerformanceA...
main.py
README.md
student_analysis_rep...
Student.csv
Problems Output Terminal Debug Console Ports
powershell
13 Pari 70.666667 C
14 Ishaan 87.333333 B
15 Myra 61.333333 D
16 Atharv 75.666667 C
17 Anvi 84.333333 B
18 Krish 48.666667 F
19 Tara 92.666667 A
Name Average Grade Result
0 Aarav 85.000000 B Pass
1 Diya 76.333333 C Pass
2 Rohan 67.666667 D Pass
3 Ananya 91.333333 A Pass
4 Vivaan 50.000000 F Pass
5 Isha 81.000000 B Pass
6 Arjun 36.666667 F Fail
7 Meera 88.666667 B Pass
8 Kabir 59.000000 F Pass
9 Saanvi 96.333333 A Pass
10 Aditya 69.666667 D Pass
11 Kiara 82.000000 B Pass
12 Reyansh 38.333333 F Fail
13 Pari 70.666667 C Pass
14 Ishaan 87.333333 B Pass
15 Myra 61.333333 D Pass
16 Atharv 75.666667 C Pass
17 Anvi 84.333333 B Pass
18 Krish 48.666667 F Pass
19 Tara 92.666667 A Pass
Top 5 Performing Students:
Name Average Grade Result
9 Saanvi 96.333333 A Pass
19 Tara 92.666667 A Pass
3 Ananya 91.333333 A Pass
7 Meera 88.666667 B Pass
14 Ishaan 87.333333 B Pass
--- Overall Class Statistics ---
Class Average: 72.13
Outline
Timeline
Ln 2, Col 19 Spaces: 4 UTF-8 CRLF { } Python Python 3.13 (64-bit) Go Live
File Edit Selection View Go Run Terminal Help
StudentPerformanceAnalytics
Sign In
Explorer
StudentPerformanceA...
main.py
README.md
student_analysis_rep...
Student.csv
Problems Output Terminal Debug Console Ports
powershell
--- Overall Class Statistics ---
Class Average: 72.13
Highest Student Average: 96.33
Lowest Student Average: 36.67
Students Passed: 18
Students Failed: 2
--- Subject-Wise Average Marks ---
Python 70.70
Statistics 72.45
Mathematics 73.25
dtype: float64
Best Performing Subject: Mathematics
Weakest Performing Subject: Python
--- Department-Wise Performance ---
Department
CSE 82.70
IT 59.22
Name: Average, dtype: float64
--- Students Who Need Improvement ---
Name Department Average Grade Result
4 Vivaan IT 50.000000 F Pass
6 Arjun IT 36.666667 F Fail
8 Kabir IT 59.000000 F Pass
12 Reyansh IT 38.333333 F Fail
18 Krish IT 48.666667 F Pass
Analysis report saved successfully!
--- Data Quality Check ---
Missing values in each column:
Student_ID 0
Name 0
Department 0
Python 0
Outline
Timeline
Ln 2, Col 19 Spaces: 4 UTF-8 CRLF { } Python Python 3.13 (64-bit) Go Live

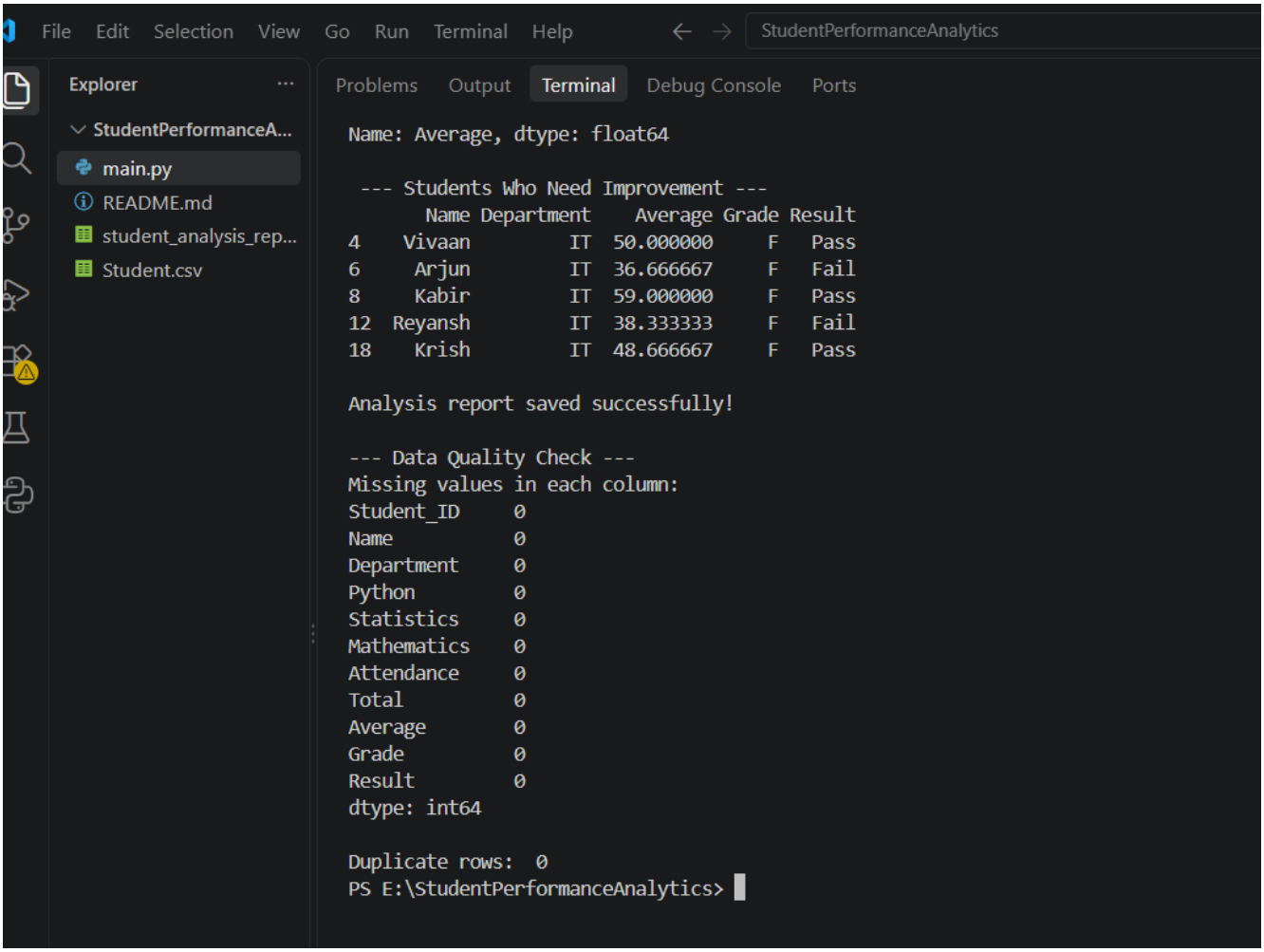
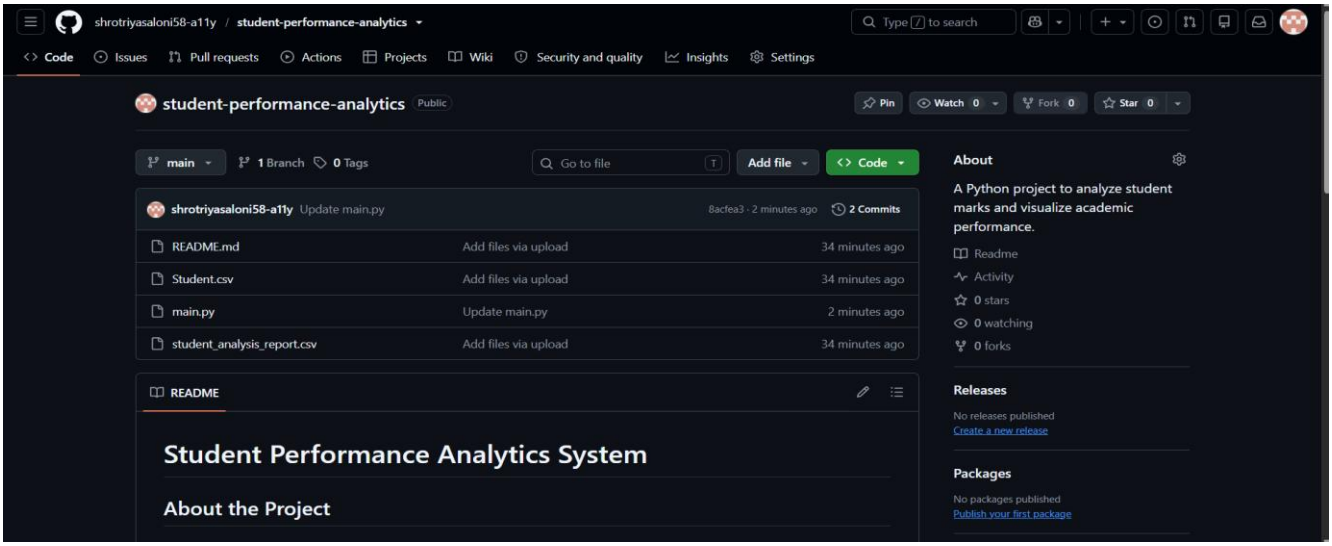


Figure 2: GitHub Repository of Student Performance Analytics System



8. Final Outcome

The Student Performance Analytics System successfully analyzes student marks using Python, Pandas, and NumPy. It calculates total and average marks, assigns grades, identifies pass or fail status, and highlights top-performing students. The system also provides subject-wise and department-wise analysis, checks data quality, exports the results to a CSV file, and displays subject averages through a bar chart. This project demonstrates how programming and data analysis libraries can be used to convert raw student data into meaningful insights.

9. Challenges & Learning

While developing this project, I learned how to handle CSV files, organize data using Pandas DataFrames, and perform numerical calculations using NumPy arrays. I gained practical experience in writing reusable functions, applying conditions, filtering and grouping records, and identifying missing or duplicate data. I also learned how to export analysis results and create visualizations using Matplotlib. Integrating these libraries helped me improve my Python programming, problem-solving, and data analysis skills.

10. GitHub Repository

https://github.com/shrotriyasaloni58-a11y/student-performance-analytics

Repository files currently listed: main.py, Student.csv, README.md, and student_analysis_report.csv.

End of Documentation